

Inference Acceleration in Large Language Models

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LLM Scaling Laws

Model performance depends most strongly on scale, which consists of three factors:

- The number of model parameters N
- The size of the dataset D
- The amount of compute C used for training

Challenges arise when scaling up

- Training difficulty
- Inference latency

Inference Acceleration via Model Design

- Mixture-of-Experts (MoE)

Activate a **subset of experts**, maintain low latency as the parameter scales up

- Diffusion LLM

Coarse-to-fine generation instead of token-by-token generation

Related Papers



Dai, Damai, Chengqi Deng, Chenggang Zhao, R. X. Xu, Huazuo Gao, Deli Chen, Jiashi Li et al. "Deepseekmoe: Towards ultimate expert specialization in mixture-of-experts language models." arXiv preprint arXiv:2401.06066 (2024).

Liu, Aixin, Bei Feng, Bin Wang, Bingxuan Wang, Bo Liu, Chenggang Zhao, Chengqi Deng et al. "Deepseek-v2: A strong, economical, and efficient mixture-of-experts language model." arXiv preprint arXiv:2405.04434 (2024).

Liu, Aixin, Bei Feng, Bing Xue, Bingxuan Wang, Bochao Wu, Chengda Lu, Chenggang Zhao et al. "Deepseek-v3 technical report." arXiv preprint arXiv:2412.19437 (2024).

Nie, Shen, Fengqi Zhu, Zebin You, Xiaolu Zhang, Jingyang Ou, Jun Hu, Jun Zhou, Yankai Lin, Ji-Rong Wen, and Chongxuan Li. "Large Language Diffusion Models." arXiv preprint arXiv:2502.09992 (2025).

Jie, Shibo, Yehui Tang, Kai Han, Yitong Li, Duyu Tang, Zhi-Hong Deng, and Yunhe Wang. "Mixture of Lookup Experts." arXiv preprint arXiv:2503.15798 (2025).

Arriola, Marianne, Aaron Gokaslan, Justin T. Chiu, Zhihan Yang, Zhixuan Qi, Jiaqi Han, Subham Sekhar Sahoo, and Volodymyr Kuleshov. "Block diffusion: Interpolating between autoregressive and diffusion language models." arXiv preprint arXiv:2503.09573 (2025).

Kaplan, Jared, Sam McCandlish, Tom Henighan, Tom B. Brown, Benjamin Chess, Rewon Child, Scott Gray, Alec Radford, Jeffrey Wu, and Dario Amodei. "Scaling laws for neural language models." arXiv preprint arXiv:2001.08361 (2020).

Inference Acceleration via Model Design

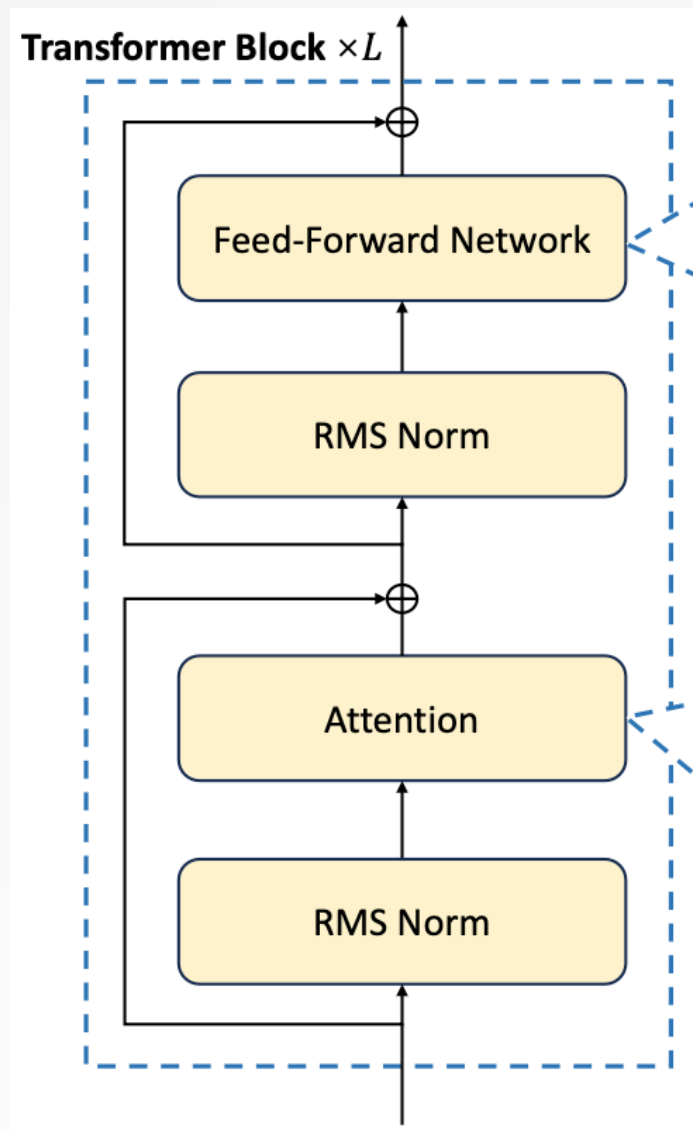
- Mixture-of-Experts (MoE)

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Mixture-of-Experts (MoE)

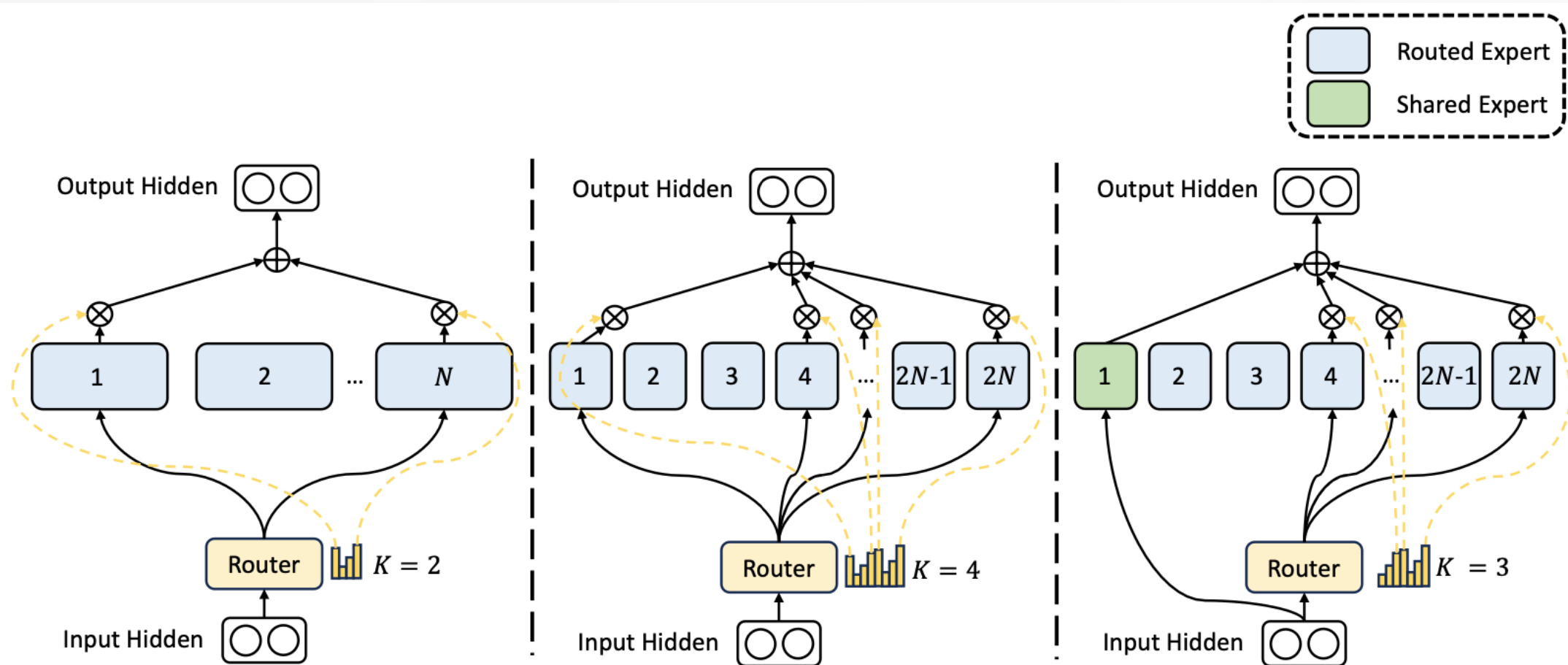


Parameter Allocation in Qwen-7B

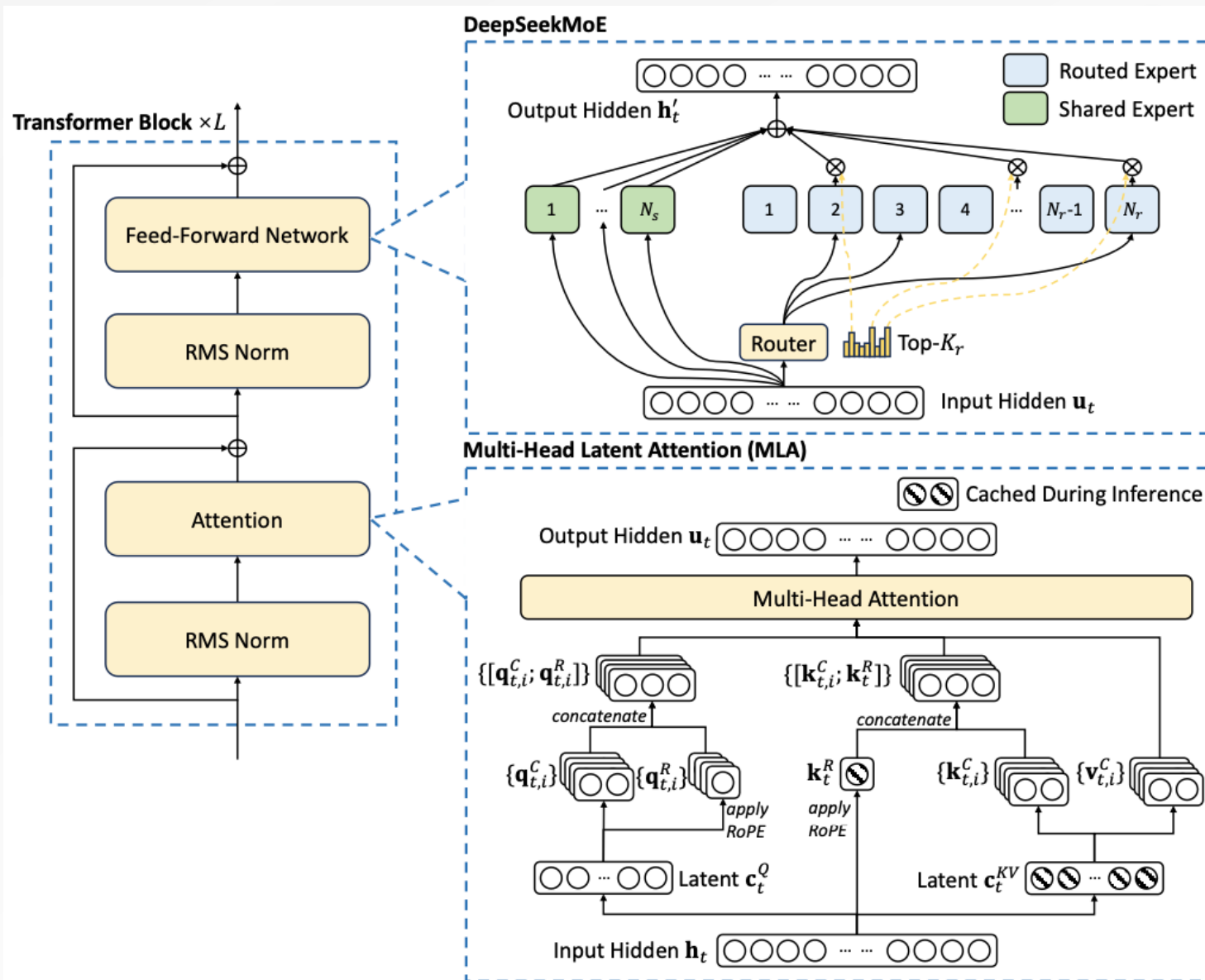
Hyperparameter	Value
Number of Layers (n_layers)	32
Number of Heads (n_heads)	32
Model Dimension (d_model)	4,096
Vocabulary Size	151,851
Sequence Length	8,192

Layer Type	Component	Number of Parameters Calculation	Percentage
Embedding		$151,936 \times 4,096 \times 2$	16.120%
LayerNorm		$4,096 \times 65$	0.003%
Attention	c_attn	$12,288 \times 4,096 \times 32 + 12,288 \times 32$	20.864%
Attention	c_proj	$4,096 \times 4,096 \times 32$	6.953%
MLP	w1+w2+c_proj	$4,096 \times 11,008 \times 2 \times 32 + 11,008 \times 4,096 \times 32$	56.059%

DeepSeekMoE



(a) Conventional Top-2 Routing → (b) + Fine-grained Expert Segmentation → (c) + Shared Expert Isolation (DeepSeekMoE)



Parameter Size Limits Deployment

- Load all experts into VRAM

Increase the memory requirement

- Load experts into VRAM only when needed

Increase inference latency

New Architecture: Mixture of Lookup Experts



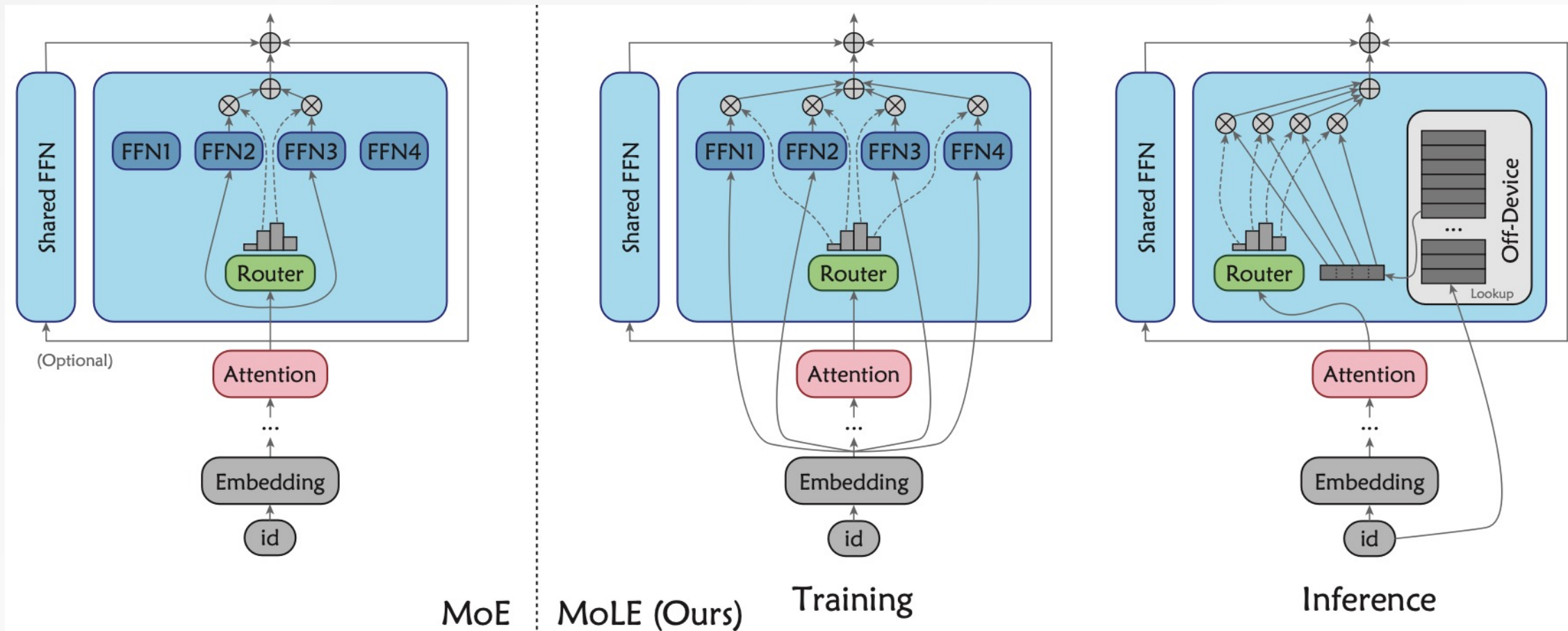
Method

- MoLE's experts are **fed with embedding tokens** (the number of tokens is finite)
- For each token, **precompute** the outputs corresponding to all experts, create lookup tables
- Directly **retrieve** the expert's outputs based on input ids and load them into VRAM

Advantages

- Computation-free experts
- Low VRAM overhead and communication latency
- Batch-generation friendly

Mixture of Lookup Experts



Results

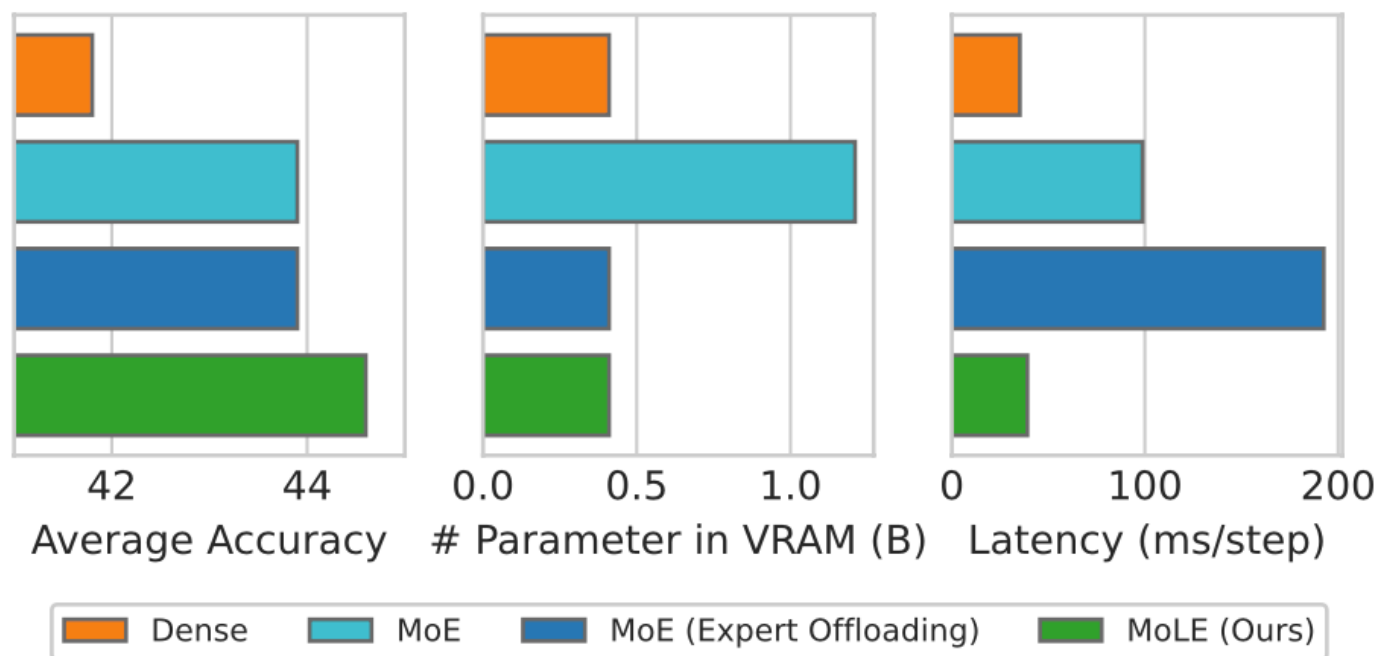


Figure 1. With the same 410M activated parameters, MoE outperforms the dense model in terms of performance, but it comes with significant VRAM usage. If experts are offloaded, inference latency will increase. Our MoLE maintains competitive performance without increasing the model's VRAM usage or decoding latency.

Inference Acceleration via Model Design

- Mixture-of-Experts (MoE)

Activate a subset of experts, maintain low latency as the parameter scales up

- Diffusion LLM

Coarse-to-fine generation instead of token-by-token generation

Large Language Diffusion Models



Autoregressive models are widely regarded as the cornerstone of LLMs. This paper challenges this by introducing a **diffusion** model trained from scratch.

LLaDA models distributions through a **forward** data **masking** process and a **reverse** process, parameterized by a vanilla Transformer to predict masked tokens.

LLaDA 8B is competitive with strong LLMs like LLaMA3 8B.

A Fundamental Question

Is the autoregressive paradigm the only viable path to achieving the intelligence exhibited by LLMs?

$$\underbrace{\max_{\theta} \mathbb{E}_{p_{\text{data}}(x)} \log p_{\theta}(x) \Leftrightarrow \min_{\theta} \text{KL}(p_{\text{data}}(x) || p_{\theta}(x))}_{\text{Generative modeling principles}}. \quad (1)$$

$$\underbrace{p_{\theta}(x) = p_{\theta}(x^1) \prod_{i=2}^L p_{\theta}(x^i | x^1, \dots, x^{i-1})}_{\text{Autoregressive formulation}}, \quad (2)$$

It is the **generative modeling principles** (Eq.(1)) rather than the autoregressive formulation (Eq.(2)), that fundamentally underpin the essential properties of LLMs.

Methodological Comparison

Challenges due to the autoregressive nature of LLMs

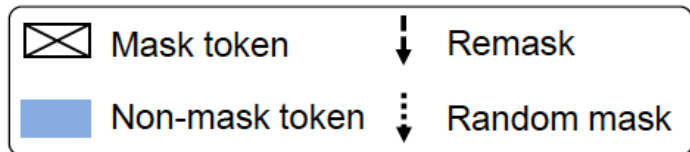
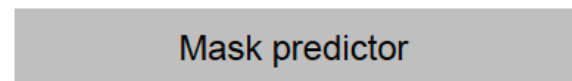
- Sequential generation incurs high computational costs
- The left-to-right modeling limits effectiveness in reversal reasoning tasks

Advantages of LLaDA

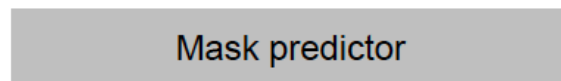
- Coarse-to-fine generation, **potential in speed**, (Mercury, the first commercial-grade diffusion large language model <https://chat.inceptionlabs.ai/>)
- Consistent performance across forward and reversal tasks
- Scales effectively up, achieves comparable results

A Conceptual Overview of LLaDA

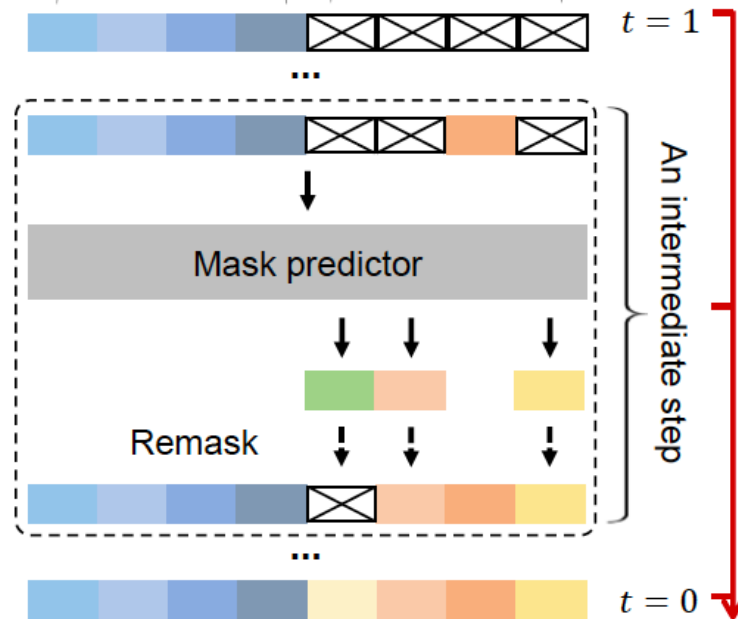
(a) Mask all tokens independently



(b) Prompt Response



(c) Prompt Response



- Pre-training. Trained on text with random masks.
- SFT. Only response tokens are possibly masked.
- Sampling. Simulates a diffusion process from $t = 1$ (fully masked) to $t = 0$ (unmasked).

Method

Model distribution $p_\theta(x_0)$: a forward process and a reverse process

- The forward process gradually **masks** tokens in x_0 until fully masked at $t = 1$.
- The reverse process **recovers** the data by iteratively predicting masked tokens as t moves from 1 to 0.

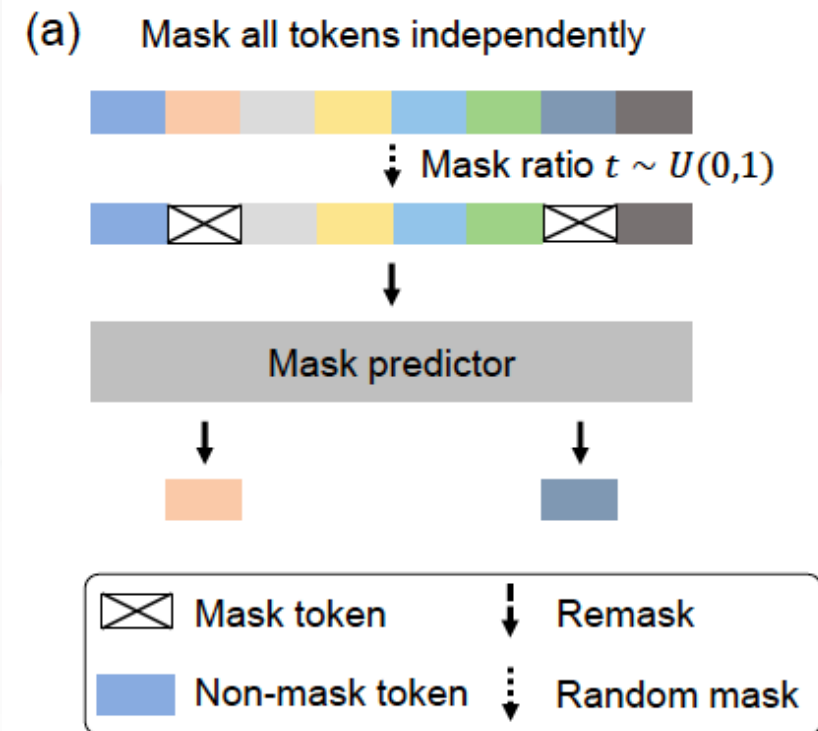
Cross-entropy loss on the masked tokens:

$$\mathcal{L}(\theta) \triangleq -\mathbb{E}_{t, x_0, x_t} \left[\frac{1}{t} \sum_{i=1}^L \mathbf{1}[x_t^i = \mathbf{M}] \log p_\theta(x_0^i | x_t) \right], \quad (3)$$

x_0 is training data, t is sampled uniformly from $[0,1]$, and x_t is sampled from the forward process.

Pre-training

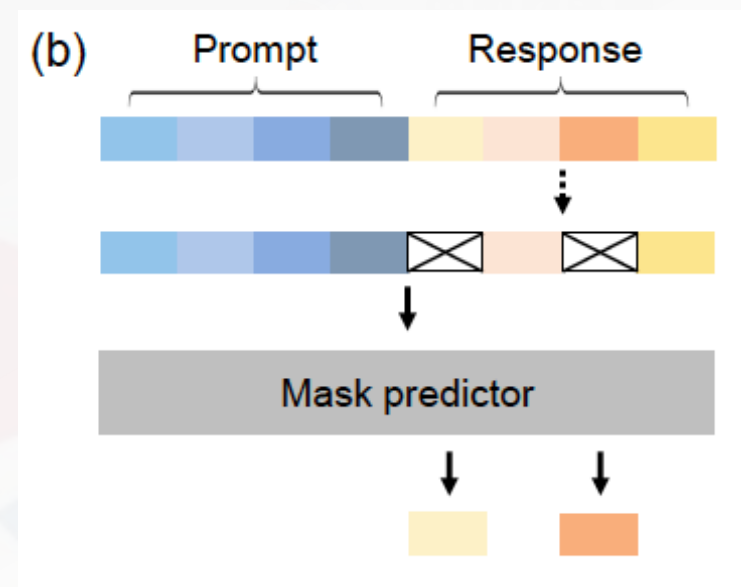
- Fixed sequence length of 4096 tokens
- Randomly sample $t \in [0,1]$ to mask each token independently with the same probability t to obtain x_t
- Dataset comprises 2.3 trillion tokens, including general text, high-quality code, math, and multilingual data.
- 1% of the pre-training data with a **random length** within $[1,4096]$, to handle variable-length data



Supervised Fine-Tuning

- Leave the prompt unchanged and **mask the tokens in the response**
- Feed both the prompt and the masked response r_t to the pre-trained mask predictor to compute the loss for SFT

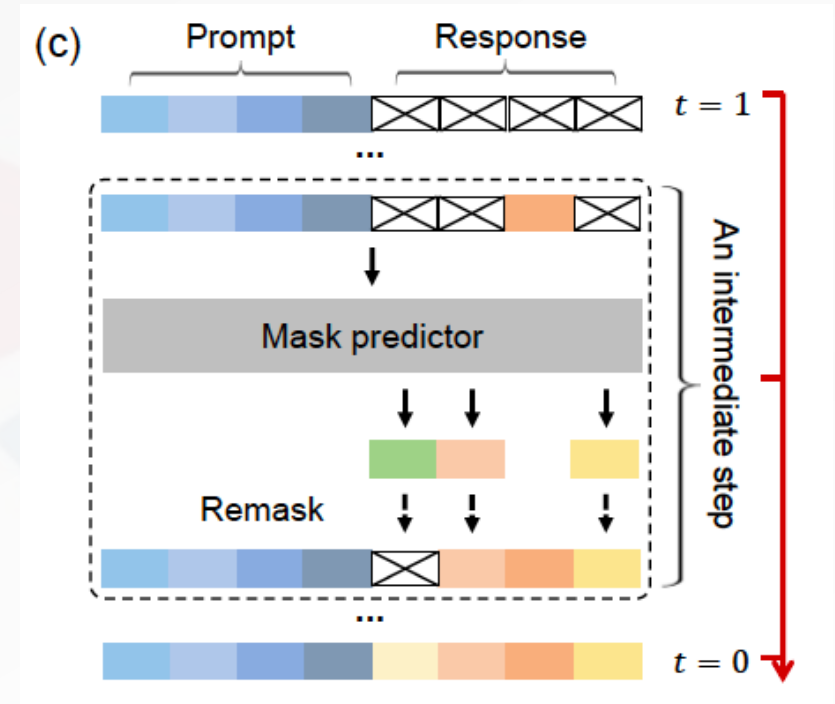
$$-\mathbb{E}_{t,p_0,r_0,r_t} \left[\frac{1}{t} \sum_{i=1}^{L'} \mathbf{1}[r_t^i = \mathbf{M}] \log p_{\theta}(r_0^i | p_0, r_t) \right], \quad (5)$$



- Dataset comprises 4.5 million pairs, including code, mathematics, instruction following, and structured data understanding

Inference

- Sample from the model distribution $p_{\theta}(r_0, p_0)$, starting from a fully masked response
- At an intermediate step from time $t \in (0,1]$ to $s \in [0, t)$, we feed both p_0 and r_t into the mask predictor and **predict all masked tokens** simultaneously
- **Remask** $\frac{s}{t}$ of the predicted tokens to obtain r_s , ensure that the transition of the reverse process aligns with the forward process
- The total number of sampling steps is a hyperparameter



Results

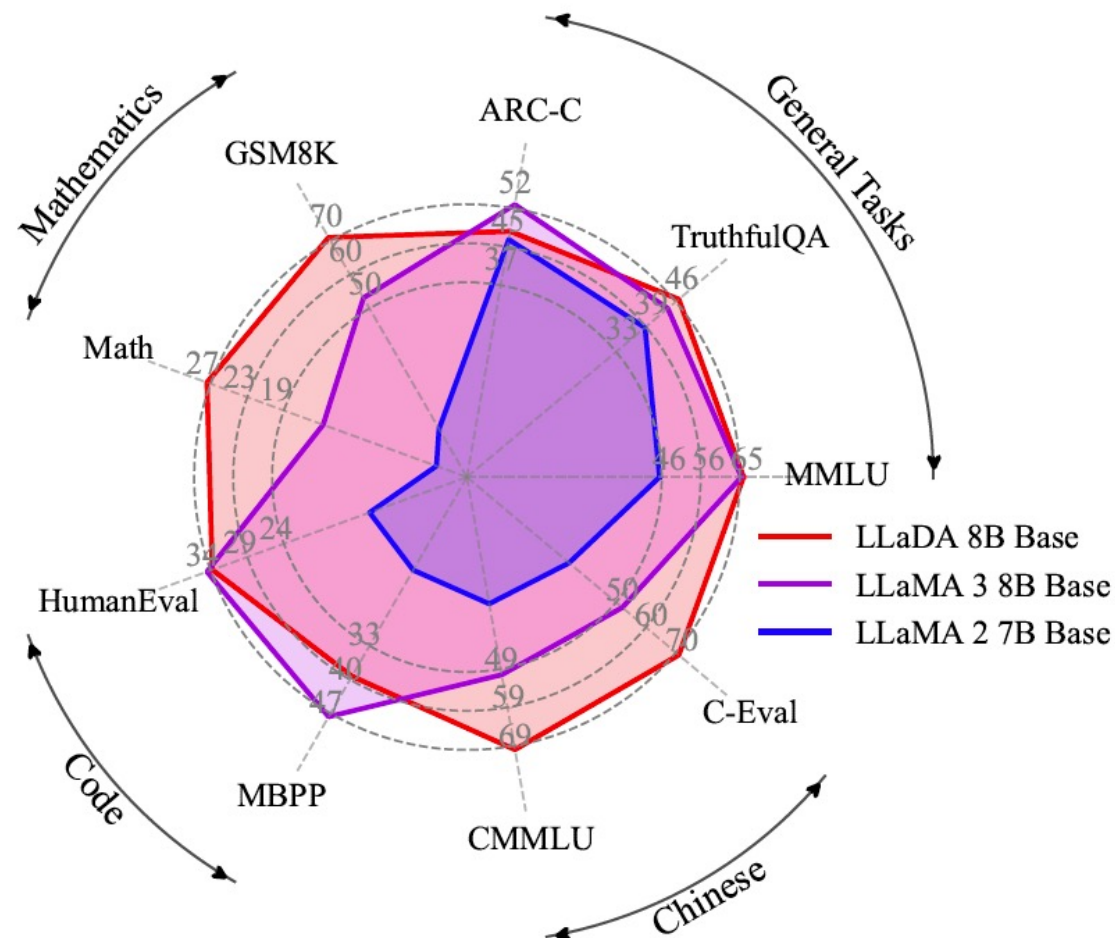


Figure 1. Zero/Few-Shot Benchmarks. We scale LLaDA to an unprecedented size of 8B parameters from scratch, achieving competitive performance with strong LLMs (Dubey et al., 2024).

Block Discrete Denoising Diffusion Language Models (BD3-LMs)

Autoregression:

✓ High quality

✓ Arbitrary-length

✓ KV caching

✗ Not Parallelizable

Generation steps

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There are three categories of the average rate
↓ There are three categories of the average rate of...

Diffusion:

✗ Lower quality

✗ Fixed-length

✗ No KV caching

✓ Parallelizable

↓ Repeal the reusability cuts and the law will continue to reduce the
the reusability will continue to the
↓ Repeal the reusability cuts and prove the law will continue to reduce the deficit.

Block Diffusion (Ours):

✓ High quality

✓ Arbitrary-length

✓ KV caching

✓ Parallelizable

↓ On September 17, we be
On September 17, 2016, we will be giving the release of
↓ On September 17, 2016, we will be giving the beta-release of the to our server testing ...

Interpolate between autoregressive and diffusion models

- Define an autoregressive distribution over blocks of tokens
- Perform diffusion within each block.



THANKS!

